



3-Terminal Positive Adjustable Regulator

LM317

Features

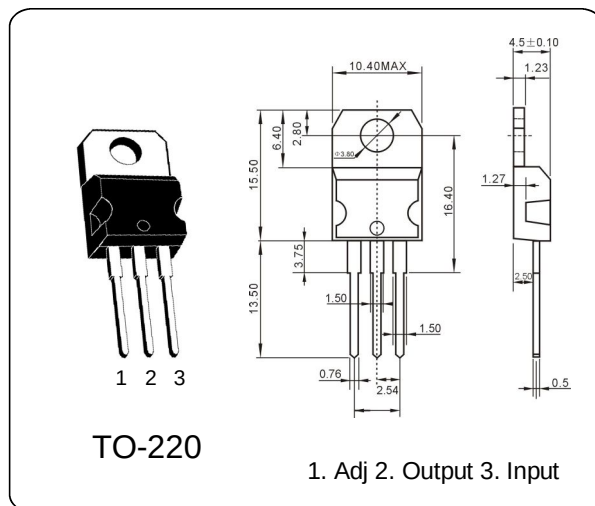
Output Current In Excess of 1. 5A
Output Adjustable Between 1. 2V and 37V
Internal Thermal Overload Protection
Internal Short Circuit Current Limiting
Output Transistor Safe Operating Area Compensation
TO-220 Package

Description

This monolithic integrated circuit is an adjustable 3-terminal positive voltage regulator designed to supply more than 1.5A of load current with an output voltage adjustable over a 1.2 to 37V. It employs internal current limiting, thermal shut-down and safe area compensation.

Absolute Maximum Ratings (Ta = 25 °C)

Parameter	Symbol	Typ	Unit
Input-Output Voltage Differential	$V_I - V_O$	40	V
Lead Temperature	T _{LEAD}	230	°C
Power Dissipation	P _D	Internally limited	W
Operating Junction Temperature Range	T _j	0~125	°C
Storage Temperature Range	T _{STG}	-65~150	°C
Temperature Coefficient of Output Voltage	V _O / T	0.02	%/°C



Electrical Characteristics (Ta = 25 °C)

($V_I - V_O = 5V$, $I_O = 0.5A$, 0°C T_j + 125°C, I_{MAX} = 1.5A, P_DMAX = 20W, unless otherwise specified)

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Line Regulation (Note1)	R _{line}	TA = +25°C 3V V _I - V _O 40V	-	0.01	0.04	% / V
		3V V _I - V _O 40V	-	0.02	0.07	
Load Regulation (Note1)	R _{load}	TA = +25°C, 10mA I _O I _{MAX} V _O < 5V V _O 5V	-	18 0.4	25 0.5	mV % / V _O
		10mA I _O I _{MAX} V _O < 5V V _O 5V	-	40 0.8	70 1.5	
Adjustable Pin Current	I _{ADJ}	-	-	46	100	A
Adjustable Pin Current Change	I _{ADJ}	3V V _I - V _O 40V 10mA I _O I _{MAX} P _D P _{MAX}	-	2.0	5	A
Reference Voltage	V _{REF}	3V V _{IN} - V _O 40V 10mA I _O I _{MAX} P _D P _{MAX}	1.20	1.25	1.30	V
Maximum Output Current	I _{O(MAX)}	V _I - V _O 15V, P _D P _{MAX} V _I - V _O 40V, P _D P _{MAX}	1.0	2.0 0.3	-	A
Ripple Rejection	RR	V _O = 10V, f = 120Hz without CADJ C _{ADJ} = 10 F (Note2)	66	60 75	-	dB

Note:

1. Load and line regulation are specified at constant junction temperature. Change in V_D due to heating effects must be taken into account separately. Pulse testing with low duty is used. (P_{MAX} = 20W)
2. C_{ADJ}, when used, is connected between the adjustment pin and ground.